VMC Vidyamandir Classes SINCE 1986

IIT JEE | MEDICAL | FOUNDATION

CODE A

JEE ADVANCED - 8 | PAPER - 1

JEE 2022

Date	Maximam Marks	Duration
13th February, 2022	180	3 Hours

General Instructions

1. The question paper consists of 3 Subject (Subject I: **Physics**, Subject II: **Chemistry**, Subject III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).

2. **Section 1** contains **2 types** of questions **[Type A and Type B].**
Type A contains **SIX (06) Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

Type B contains **SIX (06) Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

3. **Section 2** contains **THREE (03) Paragraphs.** There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)

4. For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code, Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) : ______________________

Roll Number : ______________________

OMR Bar Code Number : ______________________

Candidate's Signature : ______________ Invigilator's Signature ______________

MARKING SCHEME

SECTION-1 | Type A

- This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the answer. For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +3 If ONLY the correct option is chosen;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : –1 In all other cases.

SECTION-1 | Type B

- This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
  Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks : 0 If unanswered;
  Negative Marks : –2 In all other cases.
- For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then
  choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (D) will get +2 marks;
  choosing ONLY (A) will get +1 mark;
  choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
  choosing any other option(s) will get –2 marks.

SECTION-2

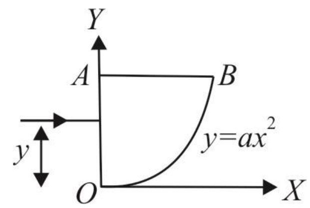
- This section contains **THREE (03)** Paragraphs. There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks: +3 If ONLY the correct Integer value is entered. There is **NO negative marking.**
  Zero Marks: 0 In all other cases.

SUBJECT I : PHYSICS 60 MARKS

SECTION-1 | Type A

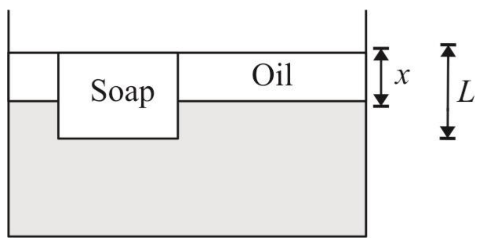
This section consists of 6 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

1. The cross-section of a prism is shown in figure. One of its refracting surfaces (OB) is given by $y = ax^2$, where a is a constant. A ray of light travelling parallel to x-axis is incident normally on face OA and refracted. The minimum distance y_{min} of incident ray from $x-axis$ to get refracted from surface OB is :



(A) $\dfrac{1}{4a(\mu^2-1)}$ **(B)** $\dfrac{\mu^2-1}{4a}$ **(C)** $\dfrac{\mu^2-1}{2a}$ **(D)** $\dfrac{1}{2a(\mu^2-1)}$

2. A rectangular bar of soap having density $800\ kg/m^3$ floats in water of density $1000\ kg/m^3$. Oil of density $300\ kg/m^3$ is slowly added, forming a layer that does not mix with the water. When the top surface of the oil is at the same level as the top surface of the soap, what is the ratio of the oil layer thickness to the soap's thickness, x/L?

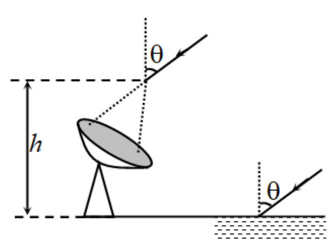


(A) $\dfrac{2}{8}$ **(B)** $\dfrac{2}{7}$ **(C)** $\dfrac{3}{10}$ **(D)** $\dfrac{3}{8}$

3. One mole of an ideal gas undergoes a process whose molar heat capacity is $4R$ and in which work done by gas for small change in temperature is give by the relation $dW = 2RdT$, then the ratio $\dfrac{C_P}{C_V}$ is :

(A) $\dfrac{7}{5}$ **(B)** $\dfrac{5}{3}$ **(C)** $\dfrac{3}{2}$ **(D)** 2

4. Radio waves of wavelength λ are received by a radar at an angle θ to vertical after reflecting from a nearby water surface and directly. If the radar records a maximum intensity, the height of antenna h from water surface can be:



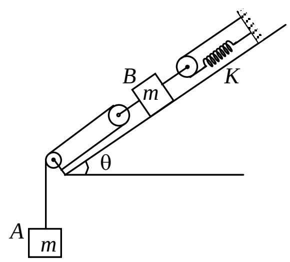
(A) $\dfrac{\lambda}{2\cos\theta}$ **(B)** $\dfrac{\lambda}{2\sin\theta}$

(C) $\dfrac{\lambda}{4\sin\theta}$ **(D)** $\dfrac{\lambda}{4\cos\theta}$

5. An isolated sphere of radius R_1 is enclosed by an earthed concentric sphere of radius R_2. Given $R_1 / R_2 = \left(1 - \frac{1}{n}\right)$. Find the ratio of the capacitance of this system to the capacity of isolated sphere of radius R_1.

(A) $\left(1-\frac{1}{n}\right)$ **(B)** $\frac{1}{n}$ **(C)** n **(D)** $\frac{n}{(n-1)}$

6. Two blocks *A* and *B*, each of mass *m* are connected by means of a pulley-spring system on a smooth inclined plane of inclination θ as shown in the figure. All the pulleys and spring are ideal. Now, *B* is slightly displaced from its equilibrium position and it starts to oscillate. Time period of oscillation of *B* will be: (Take m = 4 kg, K = 5 N/m, π = 3.14)



(A) 3.14 s **(B)** 6.28 s **(C)** 4.28 s **(D)** 5.14 s

SECTION-1 | Type B

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

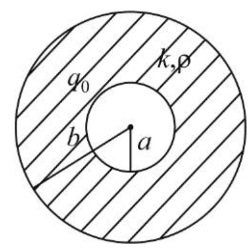
7. A non-relativistic particle of mass m is held in a circular orbit around the origin by an attractive force. The potential energy of the force is given by $U(r)=\frac{kr^2}{2}$, where r is the distance of the particle from origin and k is a constant. Assuming the Bohr quantization of the angular momentum of the particle, choose the correct statement(s:)

($v \to$ velocity of particle, $KE(r) \to$ kinetic energy of particle, $E(r) \to$ total energy of the particle)

(A) $r^2=\frac{nh}{2\pi\sqrt{km}}$ **(B)** $v^2=\frac{nh}{2\pi\sqrt{km}}$

(C) $KE(r)=U(r)$ **(D)** $E(r)=\frac{nh}{2\pi}\sqrt{\frac{k}{m}}$

8. The radii of a spherical capacitor are equal to a and $b(b>a)$. The space between the two spherical shells is filled with a dielectric of dielectric constant K and resistivity ρ. At $t=0$, the inner sphere is given a charge q_0. Choose the correct option(s):



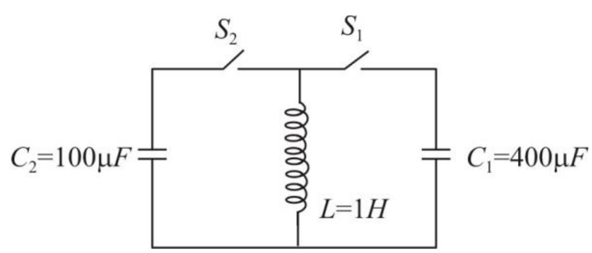
(A) Charge q on the inner sphere as a function of time is given by $q=q_0e^{-\frac{t}{\rho K\varepsilon_0}}$

(B) The charge on the inner sphere will become zero almost instantly

(C) After a long time, the charge on the outer sphere will become q_0

(D) The total amount of heat generated during the redistribution of charge will be given by $H=\left(\frac{1}{a}-\frac{1}{b}\right)\frac{q_0^2}{8\pi\varepsilon_0}$

9. In the $L-C$ circuit, the capacitor C_1 is charged to a potential of $\sqrt{2}$ volt, and the capacitor C_2 and the inductor were initially uncharged. Now switch S_1 is closed at $t=0$, and at $t=\frac{\pi}{200}$sec switch S_2 is closed and S_1 is opened. At $t=\frac{\pi}{100}$sec, S_2 is also opened.



Choose the correct statement (s):

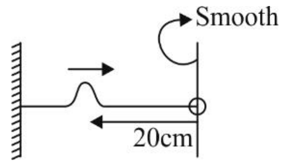
(A) When switch S_1 is opened capacitor C_1 has $400\mu C$ charge

(B) Final potential difference across capacitor C_2 is $2 Volt$

(C) Final energy of the two capacitors is equal.

(D) All the three divices have non-zero energy when switch S_2 is opened

10. A horizontal string of linear mass density $0.5g\,cm^{-1}$ and length $30\,cm$ is tied to a rigid wall at one end and to a frictionless ring at the other end. The ring can move on a vertical rod. A pulse on the string moves towards the ring at the speed of $20 cm\,s^{-1}$. The pulse is symmetric about its maximum which is initially located at $20\,cm$ from the ring end. Assume that the wave is reflected from the ends without any loss of energy. Choose the correct statement(s):



(A) The tension in the string is $2\times10^{-3}N$

(B) Minimum time taken for the string to regain its initial shape is $2s$.

(C) Minimum time for the pulse to arrive at its initial shape, location and direction of velocity is $3s$.

(D) Minimum time for the pulse to arrive at its initial shape, location and direction of velocity is $6s$.

11. Two thermometers, one containing mercury and another spirit read same temperature. The mercury thermometer has a lower emissivity than spirit thermometer. Both have the same area and heat capacity. If both are brought in bright sun.

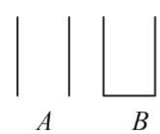
(A) The temperature rises at equal rate in both.

(B) The temperature rises at higher rate in spirit thermometer.

(C) Final steady state temperature will be the same in both

(D) Final steady state temperature will be higher in spirit thermometer.

12. Two narrow cylindrical pipes A and B are filled with hydrogen and helium gases respectively at same temperature. Pipe A is open at both ends and pipe B is open at one end and closed at the other end. If frequency of the second overtone of pipe A is equal to the frequency of the first overtone of the pipe *B*,



(A) ratio of the velocities of sound waves in the two pipes $\frac{v_A}{v_B}=\sqrt{2}$

(B) ratio of the velocities of sound waves in the two pipes $\frac{v_A}{v_B}=\sqrt{\frac{42}{25}}$

(C) ratio of their lengths $\frac{l_A}{l_B}=\sqrt{\frac{168}{25}}$

(D) ratio of their lengths $\frac{l_A}{l_B}=2\sqrt{2}$

SECTION-2

This section consists of THREE (03) Paragraphs. There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08).

Paragraph for Question 1-2

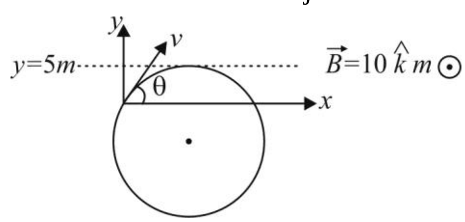
A galvanometer is used as an ammeter and a voltmeter. The range of the voltmeter can be changed to n times its original value with the help of a 27Ω multiplier. The range of the same galvanometer when used as an ammeter can be changed to n times its original value using a 3Ω shunt. Power dissipated by the galvanometer when giving a full-scale reading is $9\times10^{-4}W$.

1. The resistance of the coil (in Ω) is ________.

2. The full scale deflection current (in mA) is ________.

Paragraph for Question 3-4

A particle having mass $5\times10^{-5}kg$ and charge $1\mu c$ is projected from origin in the $x-y$ plane. Magnetic field $10\hat{k}$ is present in the region. The particle moves in a circle and just touches a line $y=5m$ at $x=5\sqrt{3}m.$

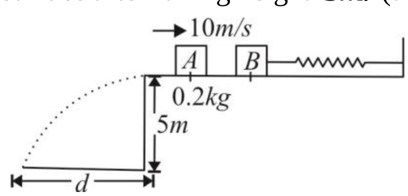


3. The angle of projection of the particle with the x-axis (in degree) is ________.

4. The speed (in m/s) with which the particle is projected is ________.

Paragraph for Question 5-6

Figure shows block A of mass $0.2kg$ sliding to the right over a frictionless elevated surface at a speed of $10m/s.$ The block undergoes a collision with stationary block $B,$ which is connected to a non deformed spring of spring constant $1000Nm^{-1}.$ The coefficient of restitution between the blocks is $0.5.$ After the collision, block B oscillates in SHM with a period of $0.2s.$ Block A eventually slides off the opposite end of the elevated surface, landing a distance $'d'$ from the base of that surface after falling height $5m.$ (use $\pi^2=10; g=10m/s^2$)



5. Amplitude of the SHM as being executed by block B-spring system is $2.5\sqrt{x}$ cm, where x is ________.

6. The distance $'d'$ (in m) will be equal to ________.

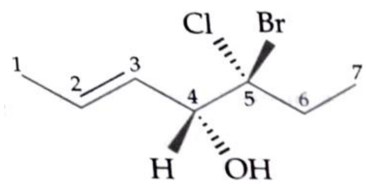
SPACE FOR ROUGH WORK

SUBJECT II : CHEMISTRY 60 MARKS

SECTION-1 | Type A

This section consists of 6 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

1. For the structure



Which of the following gives the correct designation of the configurations?

(A) 2E, 4R, 5R **(B)** 2E, 4R, 5S **(D)** 2Z, 4R, 5S **(D)** 2E, 4S, 5S

2. Identify acidic refractory material used in metallurgy :

(A) Silica **(B)** Lime **(D)** dolomite **(D)** bone ash

3. The % of ortho Hydrogen in ordinary hydrogen at room temperature is :

(A) 50% **(B)** 75% **(D)** 25% **(D)** 51%

4. Which of the following is used as a source of O_2 in space capsule ?

(A) KO_2 **(B)** K_2O_2 **(D)** KOH **(D)** Li_2O

5. 0.1 molal solution of potassium ferrocyanide is 10% dissociated at room temperature. Calculate Boiling point of this aq solution. $K_b = 0.513°C/m$, $T_{B,H_2O} = 100°C$

(A) 100.72°C **(B)** 101°C **(C)** 100.072°C **(D)** 99.23°C

6. The number of third nearest neighbours in face-centred cubic lattice is :

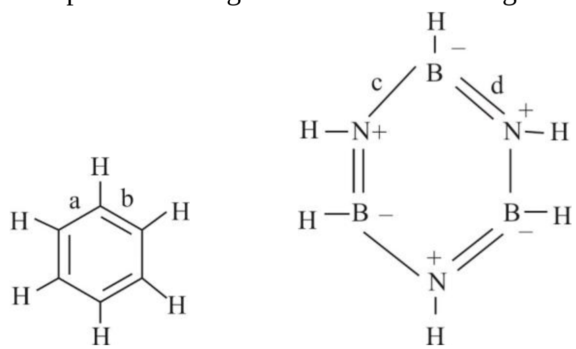
(A) 6 **(B)** 12 **(C)** 24 **(D)** 8

SPACE FOR ROUGH WORK

SECTION-1 | Type B

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

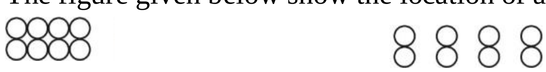
7. Compare bond length in benzene and inorganic benzene



(A) bond length $a = b$ **(B)** bond length $c = d$

(C) bond length $a < d$ **(D)** bond length $a = c$

8. The figure given below show the location of atoms in three crystallographic planes. Identify the structure.



(A) SC **(B)** FCC **(C)** BCC **(D)** B and C

9. During electrolysis of an aqueous solution of $CuSO_4$ using copper electrodes. 0.635 g of Cu is deposited at cathode then choose correct statements $(Cu = 63.5)$:

(A) Molarity of solution will remain same

(B) Molarity of $CuSO_4$ will decrease.

(C) 0.02 faradays of electricity are involved in this process

(D) 0.224L of H_2 gas is formed at Anode

10. Number of geometrical isomers of $\left[Co(NH_3)_3(NO_2)_3\right]$ are :

(A) 4 **(B)** 2 **(C)** 3 **(D)** all are optically active

11. Which of the following will have lowest boiling point.

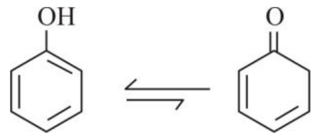
(A) $0.1M$ $Al_2(SO_4)_3$ (90% dissociated)

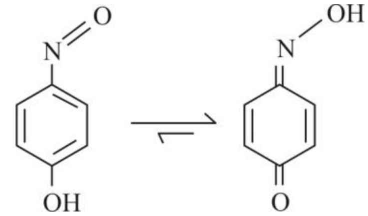
(B) $0.1M$ $BaCl_2$ (50% dissociated)

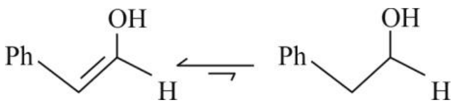
(C) $0.05M$ Na_2SO_4 (100% dissociated)

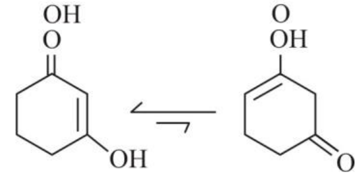
(D) $0.15M$ urea

12. Which of the following options are correctly matched in tautomerism :

(A) 

(B) 

(C) 

(D) 

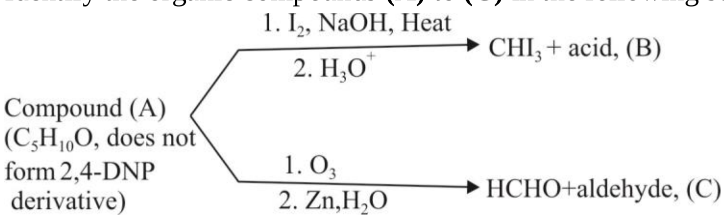
SPACE FOR ROUGH WORK

SECTION-2

This section consists of THREE (03) Paragraphs. There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08).

Paragraph for Question 1-2

Identify the organic compounds **(A)** to **(C)** in the following sequences of reaction



1. degree of unsaturation in compound B is ___________

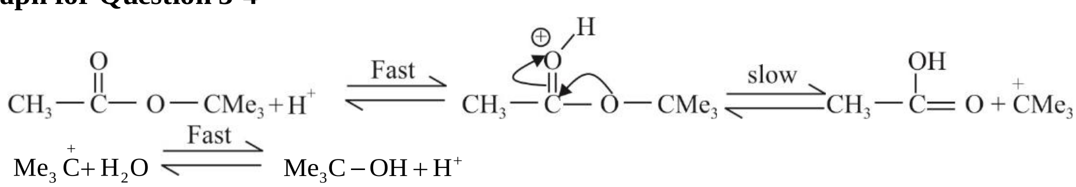
2. Consider compound C and find X.Y

Where X = Number of chiral center

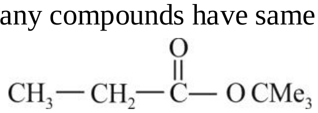
Y = Number of sp^3 hybridized carbon atoms

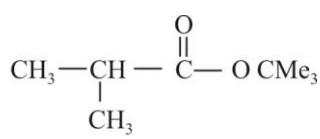
If $X = 5$, and $Y = 7$ report yours answer as 5.7

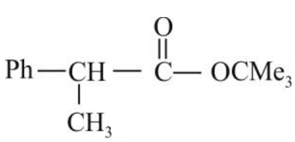
Paragraph for Question 3-4



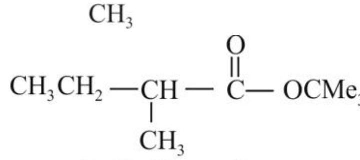
3. How many compounds have same ease of hydrolysis

I. 

II. 

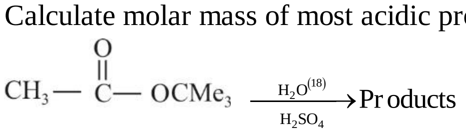
III. 

(optically pure)

IV. 

(optically pure)

4. Calculate molar mass of most acidic product



Paragraph for Question 5-6

Read the following passage and answer the questions:

When the temperature is increased, heat is supplied which increases the kinetic energy of the reacting molecules. This shall increase the number of collisions and ultimately the rate of reaction shall be enhanced. Arrhenius suggested a equation which describes k as a function of temperature, i.e.

$k = Ae^{-E_a/RT}$

Where

$k =$ rate constant

$A = a$ constant (frequency factor)

$E_a =$ energy of activation

$$\log_{10} k = \log_{10} A - \frac{E_a}{2.303R}\left[\frac{1}{T}\right]$$

$$Y = C + MX$$

It is the equation of straight line with negative slope. On plotting $\log_{10} k$ against $\left[\frac{1}{T}\right]$ we get a straight line:

5. The rate constant of a reaction is increased by 10% when its temperature is raised form $27°C$ to $28°C$. The activation energy of the reaction in kJ/mol (nearest integer) is :

$R = 8.3 J/mole\ K;\ \ln 11 = 2.4$

6. The activation energy of a second order reaction is $8.7\ kcal/mol$. The % increase in rate constant when its temperature is raised from 290 to $300K$ is approximately.

$(\log 6 = 0.78)$, $(\log_e x = 2.30 x \log_{10} x)$

SPACE FOR ROUGH WORK

SUBJECT III : MATHEMATICS 60 MARKS

SECTION-1 | Type A

This section consists of 6 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

1. If the chord of contact of tangents from two points (x_1, y_1) and (x_2, y_1) to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ are at right angles, then $\frac{x_1x_2}{y_1y_2}$ is equal to :

(A) $-\frac{a^2}{b^2}$ (B) $-\frac{b^2}{a^2}$ (C) $-\frac{b^4}{a^4}$ (D) $-\frac{a^4}{b^4}$

2. If $f(x):[1,3]\to[-1,1]$ satisfies $\int_1^3 f(x)\,dx=0$, then the maximum value of $\int_1^3 \frac{f(x)}{x}dx$ is :

(A) $\ln\left(\frac{3}{2}\right)$ **(B)** $\ln\left(\frac{4}{3}\right)$ **(C)** $\frac{8}{3}$ **(D)** $\ln\left(\frac{8}{3}\right)$

3. The value(s) of the real parameter 'a' for which the inequality $x^6-6x^5+12x^4+ax^3+12x^2-6x+1\ge 0$ is satisfied for all real x :

(A) $[-12,37]$ **(B)** $[-12,38]$ **(C)** $[-17,14]$ **(D)** $[-17,35]$

4. If $f(x)$ be a twice differentiable function such that $f(x)=x^2$ for $x=1,2,3,$ then :

(A) $f''(x)=2 \quad \forall x\in[1,3]$

(B) $f''(x)=2$ for some $x\in(1,3)$

(C) $f''(x)=2 \quad \forall x\in(1,3)$

(D) $f'(x)=2x \quad \forall x\in(1,3)$

5. $\int_0^\infty \frac{\tan^{-1}(\pi x)-\tan^{-1}x}{x}dx$ is equal to :

(A) $\frac{\pi}{2}\ln\pi$ **(B)** $\pi\ln\pi$ **(C)** $\frac{2}{\pi}\ln\pi$ **(D)** none of these

6. The domain of $f(x)=\sqrt[4]{x\left(2.3^x-\frac{4x^2+x+2}{x^2+x+1}\right)}$ is :

(A) $(-\infty, 0)\cup[101, \infty)$ **(B)** $(-\infty, 0)\cup[103, \infty)$

(C) $(-\infty, 0)\cup[93\ln 3, \infty)$ **(D)** $(-\infty,\infty)$

SECTION-1 | Type B

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

7. Which of the following statements are INCORRECT? (where {.} denotes the fractional part of function):

(A) $f(x)=\{x\}\cos^2\left(\left(\frac{2x-1}{2}\right)\pi\right)$ is differentiable for all $x \in R$

(B) $f(x)=|\sin x|\cos^{-1}(\cos x)$ is differentiable $\forall x \in (0,2\pi)$

(C) $f(x)=\big||x-2|-|x-6|\big|-3|x|+2x+1$ is not differentiable at 3 points

(D) $f(x)=\{x\}|\sin \pi x|$ is not differentiable at all integers

8. Let $f(x): R \to [-1,1]$ be twice differentiable and $f^2(0)+(f'(0))^2=4,$ then which of the following(s) is/are TRUE?

(A) There exist $\alpha, \beta \in R$ where $\alpha < \beta,$ such that f is one-one on the interval (α, β)

(B) There exists $c \in (0,2)$ such that $|f'(c)| \le 1$

(C) $\lim\limits_{x \to \infty} f(x)=1$

(D) There exists some x_0 such that $f(x_0)+f''(x_0)=0$ but $f'(x_0) \ne 0$

9. If $f(x)=2|x+1|-|x-1|+3|x-2|+2x+1$ and $h(x)=\begin{cases} 2x: & x \ge 3 \\ 1: & x < 3 \end{cases},$ then choose the correct statements :

(A) $y=h(f(x))$ is continuous $\forall x \in R$

(B) $y=h(f(x))$ is discontinuous at 3 points

(C) $y=h(f(x))$ is not differentiable at 3 points

(D) $y=h(f(x))$ is not differentiable at 4 points

10. Let $f(x)=\frac{x^2+4x+3}{x^2+7x+14}$ and $g(x)=\frac{x^2-5x+10}{x^2+5x+20},$ then :

(A) greatest value of $f(x)g(x)$ is 5

(B) greatest value of $(g(x))^{f(x)}$ is 9

(C) greatest value of $f(x)$ is 2

(D) greatest value of $(g(x))^{f(x)}$ is 8

11. Let $f:[0,1]\to R$ be a continuous function, $g(x)$ be a non-increasing function on $[0,1]$ and $h(x):[0,\infty)\to[0,\infty)$ be a continuous strictly increasing function with $h(0)=0,$ then :

(A) $\alpha\int_0^1 g(x)dx \le \int_0^\alpha g(x)dx$ for any $\alpha\in(0,1)$

(B) $\left(\int_0^1 f(t)dt\right)^2 \le \int_0^1 f^2(t)dt$

(C) $\int_0^a h(x)dx+\int_0^b h^{-1}(x)dx \ge ab$

(D) $\int_0^a h(x)dx+\int_0^b h^{-1}(x)dx \le ab$

12. Let $n\ge 0$ be an integer and $l_n=\int_0^\pi \frac{1-\cos nx}{1-\cos x}dx,$ then :

(A) $l_0,l_1,l_2,.........$ is an A.P.

(B) $l_0,l_1,l_2.......$ is a G.P.

(C) $l_n=n\pi;n\ge 1$

(D) For $n=2022$, absolute value of the integrand is bounded

SECTION-2

This section consists of THREE (03) Paragraphs. There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08).

Paragraph for Question 1-2

Let S_1,S_2,S_3 be the circles $x^2+y^2+3x+2y+1=0, x^2+y^2-x+6y+5=0$ and $x^2+y^2+5x-8y+15=0,$ then :

1. Point from which length of tangents to these three circles is same is (α,β) then $\alpha+\beta$ is ____.

2. Radius of circle S_4 which cut orthogonally to all given circles is $R,$ then $\frac{R^2}{9}$ is ________ .

Paragraph for Question 3-4

Let A be the set of all 3×3 symmetric matrices whose entries are $1,1,1,0,0,0,-1,-1,-1.$ B is one of the matrix in set A and $X=\begin{bmatrix}x\\y\\z\end{bmatrix}$ $U=\begin{bmatrix}0\\0\\0\end{bmatrix}$

3. Number of such matrices B in set A is λ, then λ is __________

4. Number of matrices B such that equation $BX = U$ has infinite solutions is _______.

Paragraph for Question 5-6

Eight digit number can be formed using all the digits $1, 1, 2, 2, 3, 3, 4, 5$.

5. A number is selected at random then the probability such that no two identical digits appear together is λ, then 168λ is :

6. A number is selected at random then the probability that it has exactly two pair of identical digits occurring together is k, then $63k$ is _______.

SPACE FOR ROUGH WORK

End of JEE Advanced-8 | PAPER-1 | JEE-2022